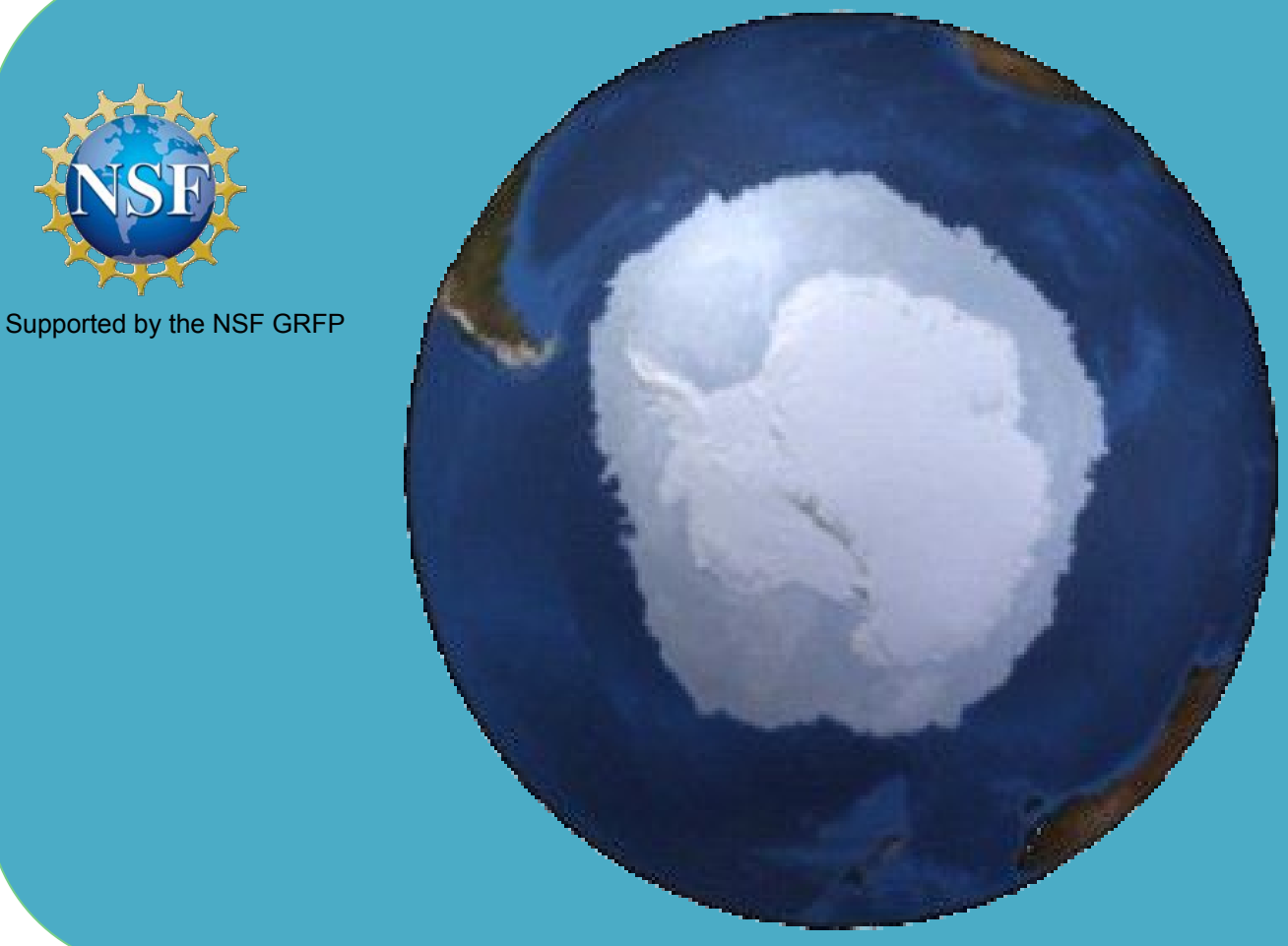




Do Climate Models Simulate the Right Sea Ice Trends for the Wrong Reasons?

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1. Motivation

CMIP5 models struggle to simulate the observed Arctic and Antarctic sea ice retreat (Figure 1b-c). Several recent studies suggest that, because the observations still fall within the range of simulated sea ice trends in each hemisphere, this disparity can be explained by natural variability (see Section 3). However, **a model may simulate a sea ice trend that is consistent with the observations for the wrong reasons, such as if a model only simulates an accurate Arctic sea ice retreat while simulating far too much global warming** (see Section 2). This concern is noteworthy because the CMIP5 models tend to simulate too large a rate of global warming during 1979-2013 (Figure 1a).

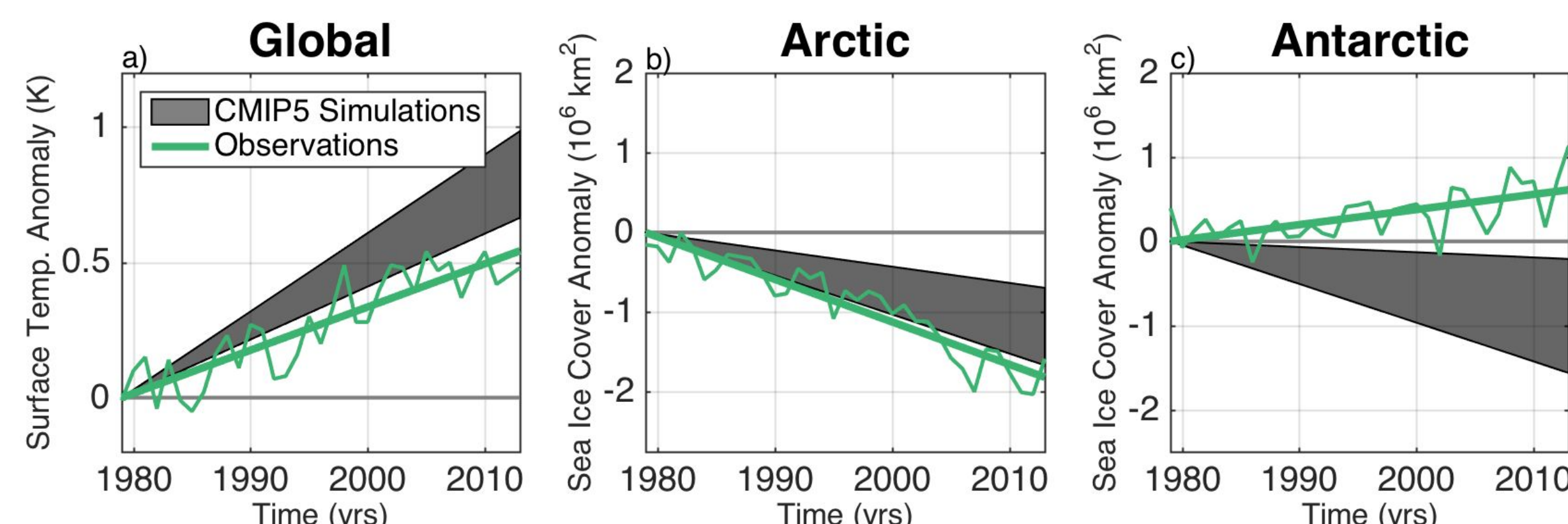


Fig. 1: Observed and CMIP5 modeled annual-mean (a) global-mean surface temperature trends, (b) Arctic, and (c) Antarctic sea ice area trends. Shadings indicate one standard deviation around the CMIP5 ensemble mean.

2. Sea ice sensitivity to global warming

Note: In this section, 30 simulations from the CESM-LE ensemble are examined. Each simulation comes from a single GCM, NCAR CESM2.0, and the simulations differ due to simulated natural variability alone.

Simulated global surface temperature and sea ice cover in each hemisphere is linearly related in each of the 30 CESM-LE model simulations of 1920-2080 (correlation of 0.99 and 0.97 in the Arctic and Antarctic, respectively).

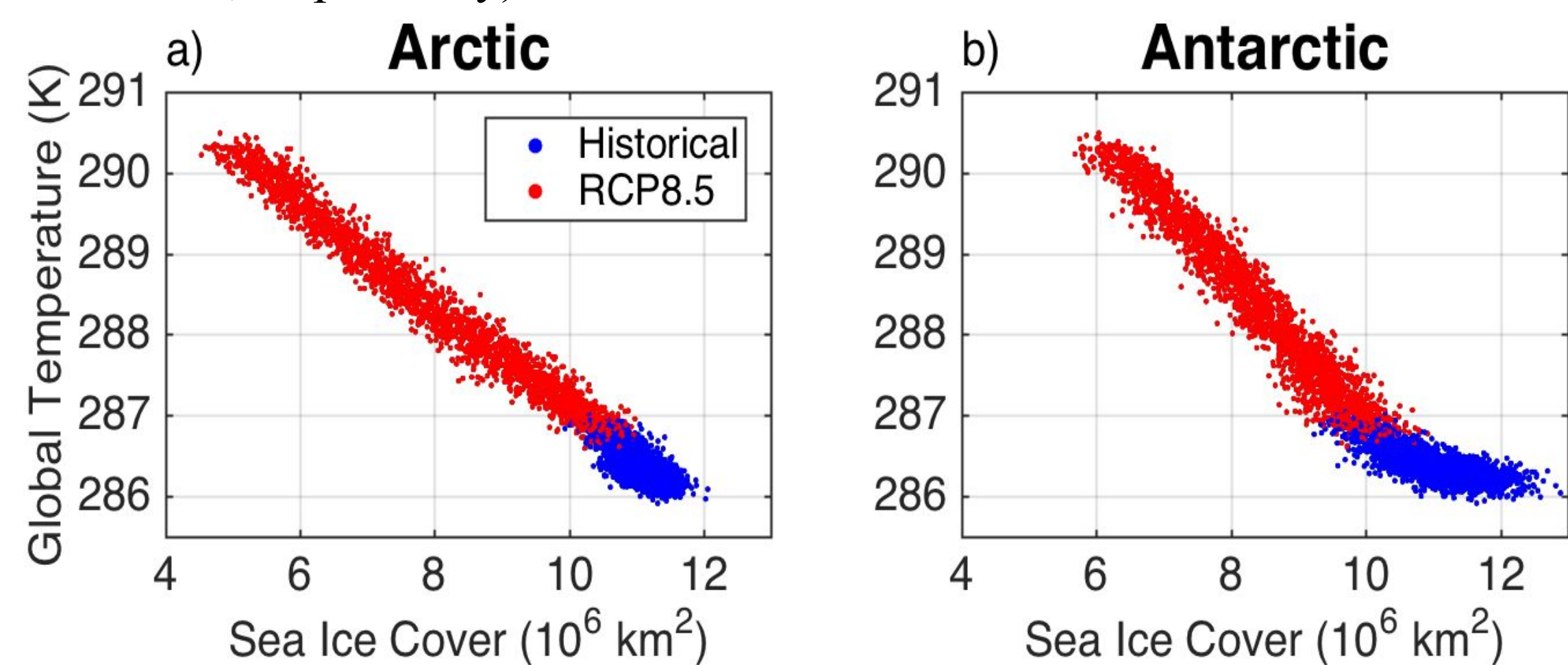


Fig. 2: CESM-LE simulated annual-mean (a) Arctic and (b) Antarctic sea ice cover versus annual-mean global-mean surface temperature associated with Historical (blue) and RCP8.5 (red) forcing scenarios during 1920-2080.

The linearity of this relationship implies that errors in global warming may lead to errors in sea ice changes (Winton 2011). This possibility is examined for sea ice changes that occur over 35-year periods (as in 1979-2013) **using all overlapping 35-year periods in the CESM-LE simulations of 1920-2080**. A relationship is visually apparent (especially in the Arctic), implying that much of the variation in sea ice trends is associated with specific changes in the level of global warming.

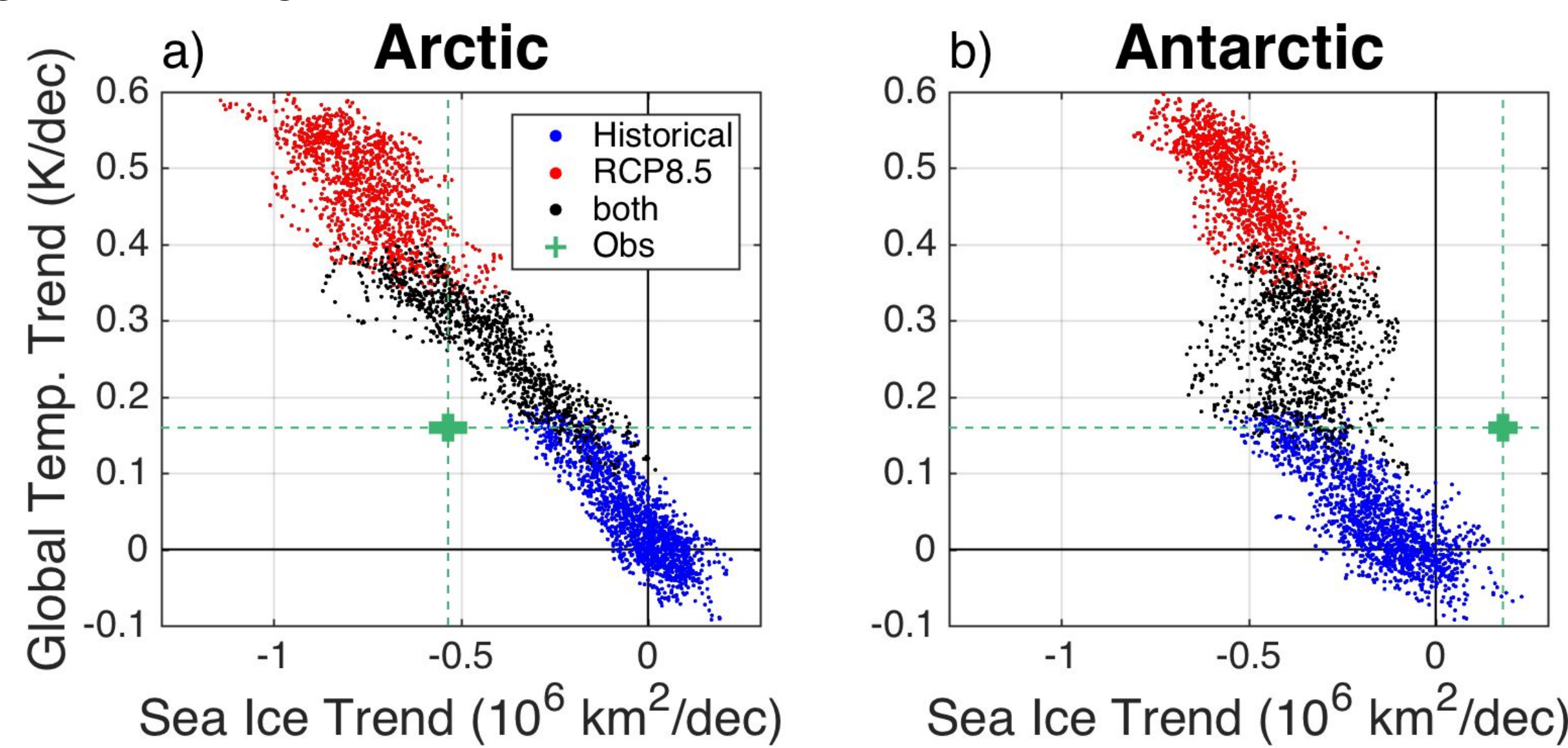


Fig. 3: Observed and simulated annual-mean (a) Arctic and (b) Antarctic sea ice trends plotted versus the corresponding annual-mean global-mean surface temperature trends from all overlapping 35-year periods in the 30 CESM-LE simulations of 1920-2080 are plotted (3,810 points in total). Green crosses represent 68% confidence intervals around the 1979-2013 observations.

This relationship is also evident in the 30 simulations of 1979-2013.

→ Realizations of natural variability that have anomalously large levels of global warming during 1979-2013 also tend to have anomalously large levels of sea ice retreat.

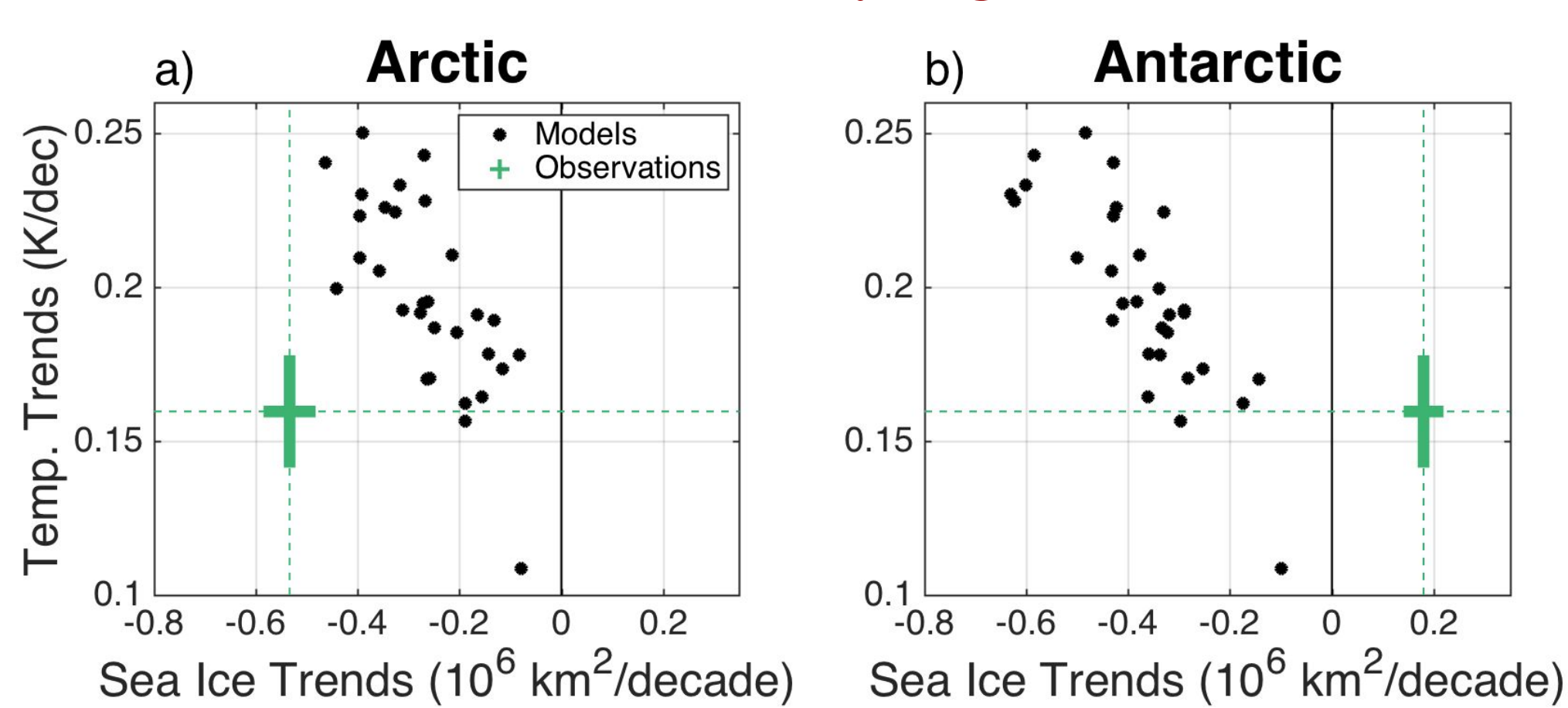


Fig. 4: 1979-2013 observed and simulated annual-mean (a) Arctic and (b) Antarctic sea ice trends plotted versus the annual-mean global-mean surface temperature trends from 30 CESM-LE simulations. Green crosses represent 68% confidence intervals around the observations.

3. Can natural variability explain the difference between observed and modeled sea ice trends?

Note: In this section, simulations from the CMIP5 ensemble of years 1979-2013 are examined (this is an ensemble of GCMs used for simulating current and future climate change in the IPCC AR5). There are 37 different climate models developed at international modeling centers, many of which submitted multiple simulations, leading to a total of 101 ensemble members. Therefore, the CMIP5 ensemble members differ due to both inter-model differences and natural variability.

The spread among the simulations represents a range of possible realizations of natural variability in the climate system, as well as inter-model differences. Therefore an individual simulation would not typically match the observations on decadal timescales even if the model were perfect, but the observations are expected to fall within the range of the simulations.

→ Recent studies suggest that simulated natural variability can explain the discrepancy between the CMIP5 ensemble mean and the observations (Fig 5a-b).

Biases in global warming influence associated sea ice trends.

(On average, simulations of 1900-2100 have a correlation between global surface temperature and sea ice cover that is -0.94 in the Arctic and -0.85 in the Antarctic.)

- All simulations with accurate Arctic sea ice trends have global warming trends that are 1.5-2 times larger than the observed trend (Figure 5c).
- Accurately simulated temperature trends appear to be associated with Antarctic sea ice retreat rather than the observed expansion (Figure 5d).

Global warming biases can approximately be accounted for by normalizing each simulated sea ice trend:

$$\left(\frac{dI}{dt}\right)_{\text{eff}} \equiv \left(\frac{dT}{dt}\right)_{\text{sim}} \left(\frac{dT}{dt}\right)_{\text{obs}}$$

The “effective sea ice trend” (crudely) estimates the magnitude of the simulated sea ice trend if the model accurately captured the observed level of global warming (Figure 5e-f).

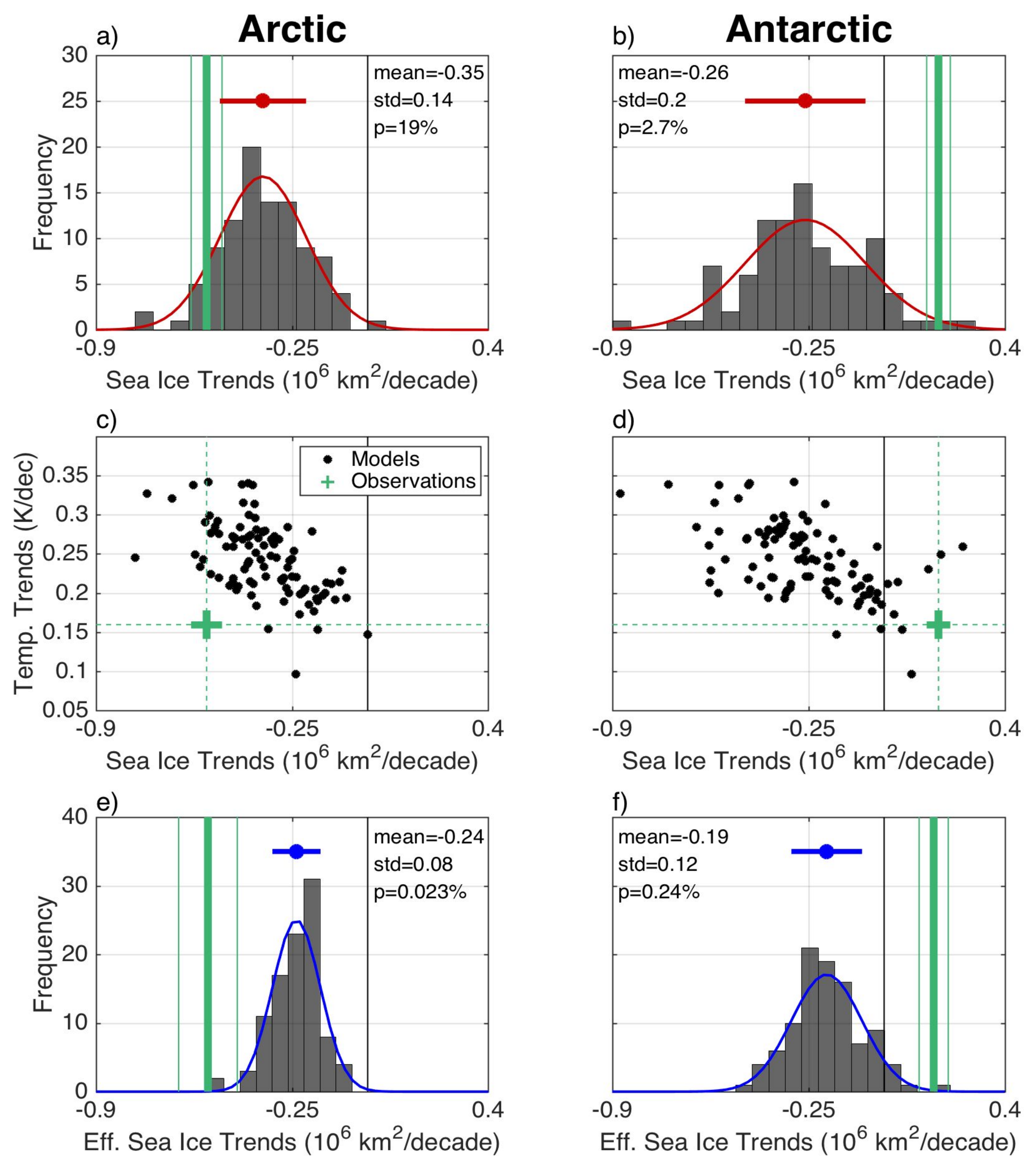


Fig 5. Observed and CMIP5 modeled annual-mean (a) Arctic sea ice trends and (b) Antarctic sea ice trends. Each histogram represents the distribution of simulated trends from 101 CMIP5 simulations, with the observed trend indicated by a thick green line, with surrounding thinner lines indicating 1-sigma confidence intervals in each panel. Standard deviations of the distribution around the CMIP5 ensemble mean are also shown. Observed and CMIP5 simulated annual-mean (c) Arctic and (d) Antarctic sea ice trends plotted versus the corresponding annual-mean global-mean surface temperature trends. Dashed lines represent the observed trends, and crosses represent the 1-sigma confidence interval. (e-f) As in top row, but using effective sea ice trends

→ Accounting for global warming biases causes sea ice trend distribution to change substantially (compare top and bottom panels).

- (1) The predicted ensemble-mean sea ice retreat becomes slower in each hemisphere (by approximately 25-30%).
→ the sea ice response to external forcing is smaller under the observed level of global warming.
- (2) The width of the simulated sea ice trend distribution decreases (the standard deviation decreases by approximately 40%).
→ for a given temperature trend, there is a much narrower distribution of sea ice trends that can arise due to natural variability.
- (3) The bias between the models and the observations is larger in the Arctic than in the Antarctic.
- (4) **If the models are correct, then the probability that natural variability would allow the observed sea ice trends to fall this far from the ensemble mean drops from 19% and 2.7% in the Arctic and Antarctic, respectively, to 0.023% and 0.24%**

4. Can climate models simulate asymmetric sea ice trends?

By indicating the extent to which each simulation over- or under-predicts the global-mean surface temperature trend, Figure 6 illustrates that **simulations that warm too rapidly compared to observations are more likely to accurately predict Arctic sea ice trends but simulate too much sea ice retreat in the Antarctic, and vice versa.**

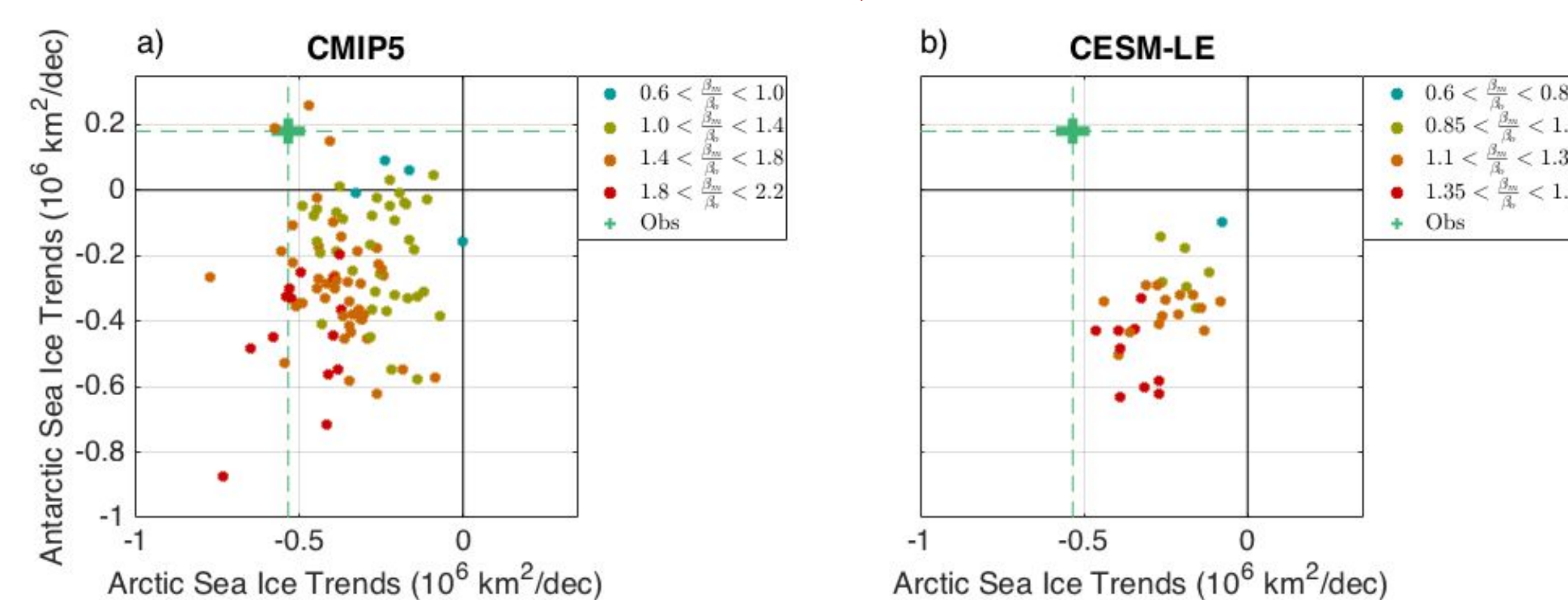


Fig 6: Antarctic sea ice trends versus the Arctic sea ice trends. Green dashed lines represent the observed trends, and the color of each simulation indicates the ratio between the modeled (m) and observed (o) values of the annual-mean global-mean surface temperature trend, using (a) CMIP5 and (b) CESM-LE model output. Green crosses indicate 68% confidence intervals around observations.

5. Conclusions

- 1) Arctic and Antarctic sea ice cover are closely related to the global-mean surface temperature in most GCMs
→ Errors in sea ice trends may arise from errors in global warming
→ Most simulations do not capture the observed asymmetry between Arctic and Antarctic sea ice trends
- 2) The “effective sea ice trend” can account for biases in global warming
- 3) We estimate that, under the observed level of global warming, the possibility that natural variability alone might explain the difference between models and observations in either hemisphere becomes exceedingly small.
→ **This implies that important processes are likely missing or are poorly simulated in the models.**

References

Winton, M. Do Climate Models Underestimate the Sensitivity of Northern Hemisphere Sea Ice Cover? Journal of Climate 24 , 3924–3934 (2011).